

CSE 311:

Data Communication

Instructor:

Dr. Md. Monirul Islam

Course Outline

- Introduction to digital **data communication**;
- Introduction to **signals**
- Review of **Fourier Transform**
- Frequency Response of **Linear Systems**;
- **Sampling** theorem;
- **Quantization**;
- **Line coding**;
- Analog **Modulation**
- Pulse modulation
- **Intersymbol interference**;
- Pulse shaping;
- **Digital modulation**
- **Multiple access techniques**:
TDM, FDM; Random processes;
- Additive White Gaussian Noise (AWGN);
- Error rate due to noise
- Introduction to **information theory**; Concept of channel coding and capacity.

Course Objectives

- Learn continuous and discrete-time signals, linear systems and their properties
- Learn and apply Fourier and DFT for analyzing signals and linear systems,
- Investigate the concepts of modulation, multiplexing and multiple access
- have in-depth knowledge and understanding of Data Comm. Techniques
- identify and compare pros and cons of different CoE techniques

Course Outcomes

CO No.	CO Statement After undergoing this course, students should be able to:	Corresponding PO(s)*	Domains and Taxonomy level(s)**	Delivery Method(s) and Activity(-ies)	Assessment Tool(s)
CO1	Understand and analyze continuous and discrete-time signals, linear systems, their properties, and the sampling process in time and frequency domains	PO1, PO2	C2, C4	Lecture	Class Tests or Assignments or Projects, and Final Exam
CO2	Apply mathematical concepts for analysis of signals and linear systems	PO1	C3	Lecture	Class Tests or Assignments or Projects, and Final Exam
CO3	Understand and analyze underlying notions of analog and digital communication systems.	-	C2, P7	Lecture	Class Tests or Assignments or Projects, and Final Exam
CO4	Evaluate and analyze different modulation, multiplexing and multiple access principals, practices and measures.	PO2, PO4, PO5	C6, A5	Lecture	Class Tests or Assignments or Projects, and Final Exam
CO5	Develop different new techniques for solutions of real-world problems	PO3, PO6, PO7	C5	Lecture	Class Tests or Assignments or Projects, and Final Exam

Text Books

1. **Modern Digital and Analog Communication Systems,**
International 4th edition, B P Lathi and Zhi Ding
2. **An Introduction to Analog and Digital Communications,**
2nd edition, S. Haykin and M. Moher

Assessment

- Class Tests: 20%
- Attendance: 10 %
- Term final: 70%

Assessment

- 2 Class Tests in the first part
 - 4th week : Monday
 - 6th week: Monday

in absence of central CT routine

Motivation

- **CoE made life easy**
- **Most of the PhD's in the Department from communication background**
 - Funding is easier
- **See: Maths are working**
- **Application to other fields**
 - Computer networks
 - Digital signal processing
 - Image/video processing
 - Computer vision

Communication systems

- Continued from ancient times
 - Runners
 - Carrier pigeons
 - Lights and mirrors
 - Smoke and fire

Communication systems

- Continued from ancient times

- Runners
- Carrier pigeons
- Lights and mirrors
- Smoke and fire

- Slow
- Unreliable

Communication systems

- Continued from ancient times
 - Runners
 - Carrier pigeons
 - Lights and mirrors
 - Smoke and fire
- Slow
- Unreliable
- However, that was enough for the time

Communication systems

- **Electrical communication systems**
 - **Reliable**
 - **dependable**
 - **economical**

Communication systems

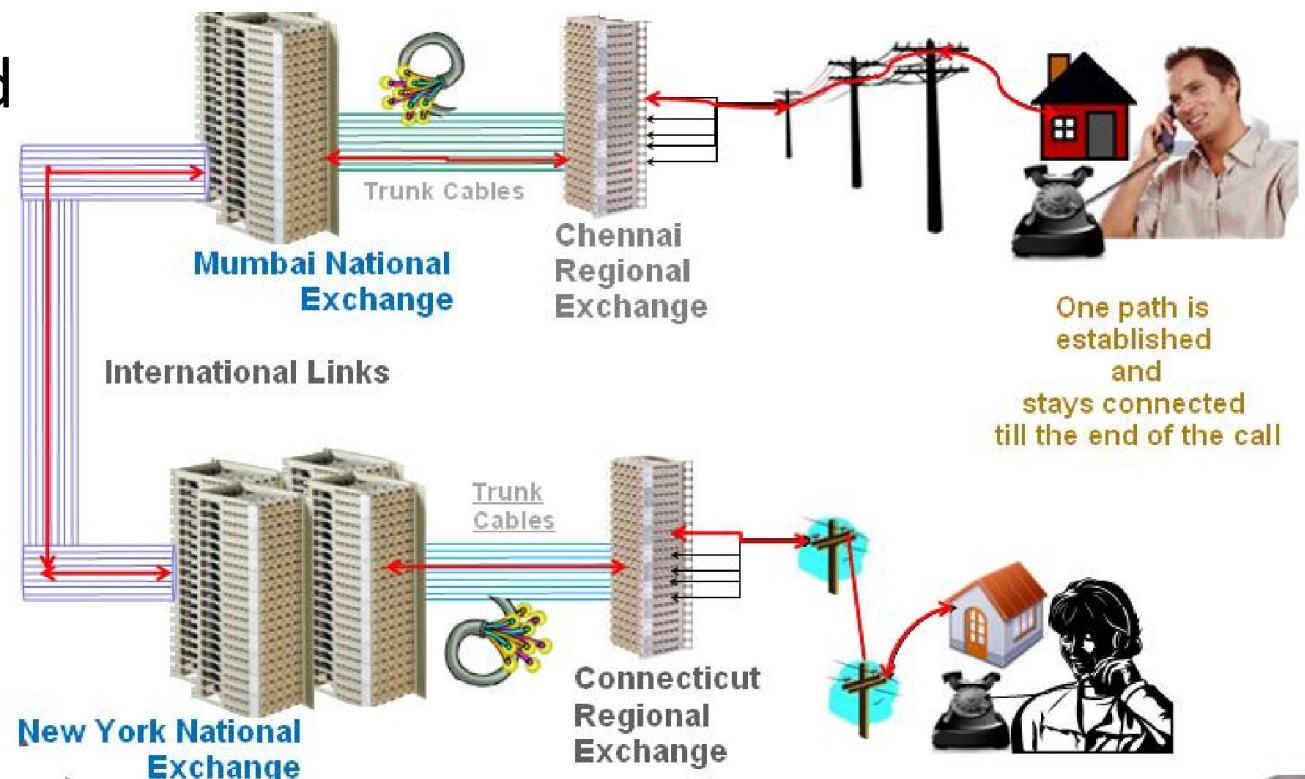
- **Electrical communication systems**
 - **Reliable**
 - **dependable**
 - **economical**
- **Email**
- **E-commerce**
- **Intercontinental business meeting, zoom, team, and many more...**

Communication system breakthroughs

- Telegraph (1844, Samuel Morse)
 - Transmitted “What hath God wrought” between Washington, D.C. and Baltimore, Maryland.

Communication system breakthroughs

- **Telephone** (1875, Alexander Graham Bell)
 - Made **real-time transmission** of speech by electrical encoding and replication of sound



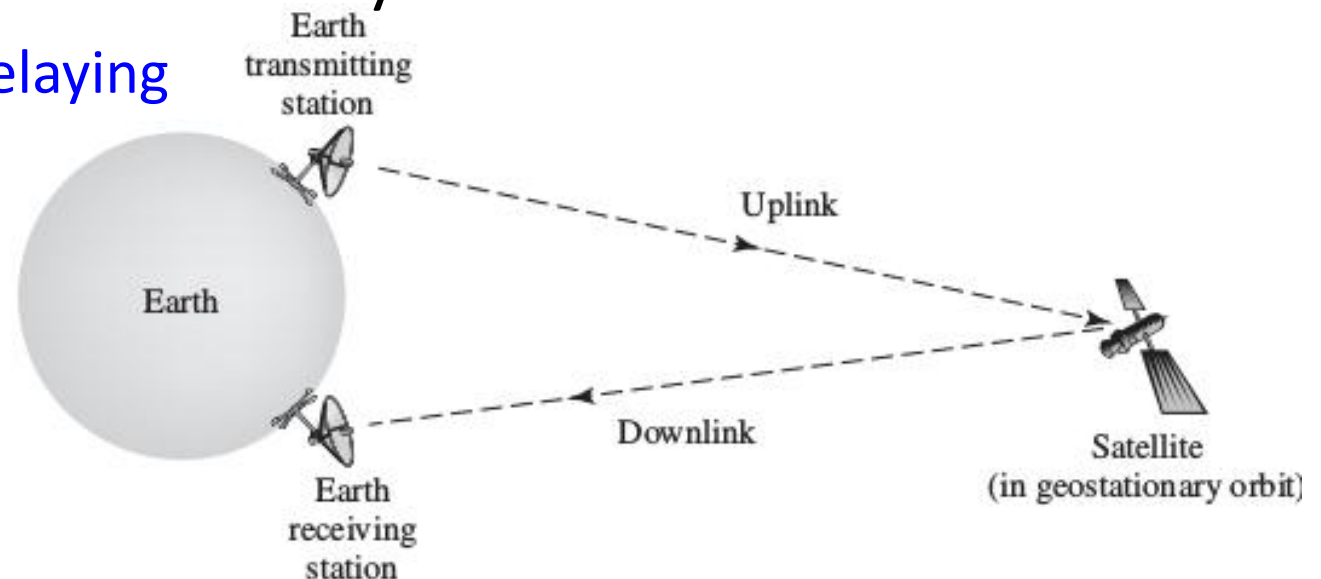
Communication system breakthroughs

- Radio (1901, Marconi) and TV (1925, Jenkins) broadcasting
- 1906:
 - AM radio
- 1935:
 - FM radio
- 1953
 - Color TV
- Recent innovation: internet radio/TV



Communication system breakthroughs

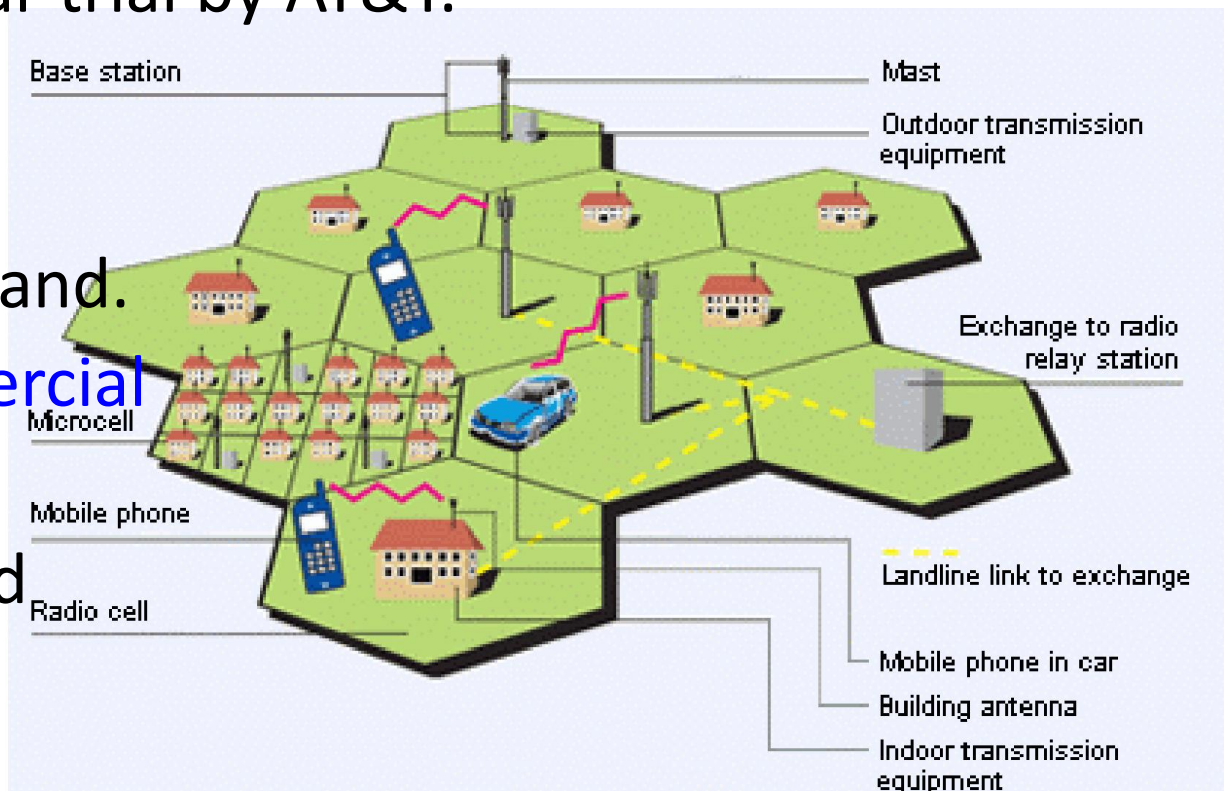
- Satellite Communication
 - 1955: John R. *Pierce proposed*
 - 1957: *Sputnik I* launched by *Soviet U* transmitted telemetry signals for 21 days.
 - 1962: *Telstar I* launched by *Bell Lab*.
 - Capable of relaying TV programs across the Atlantic.



Communication system breakthroughs

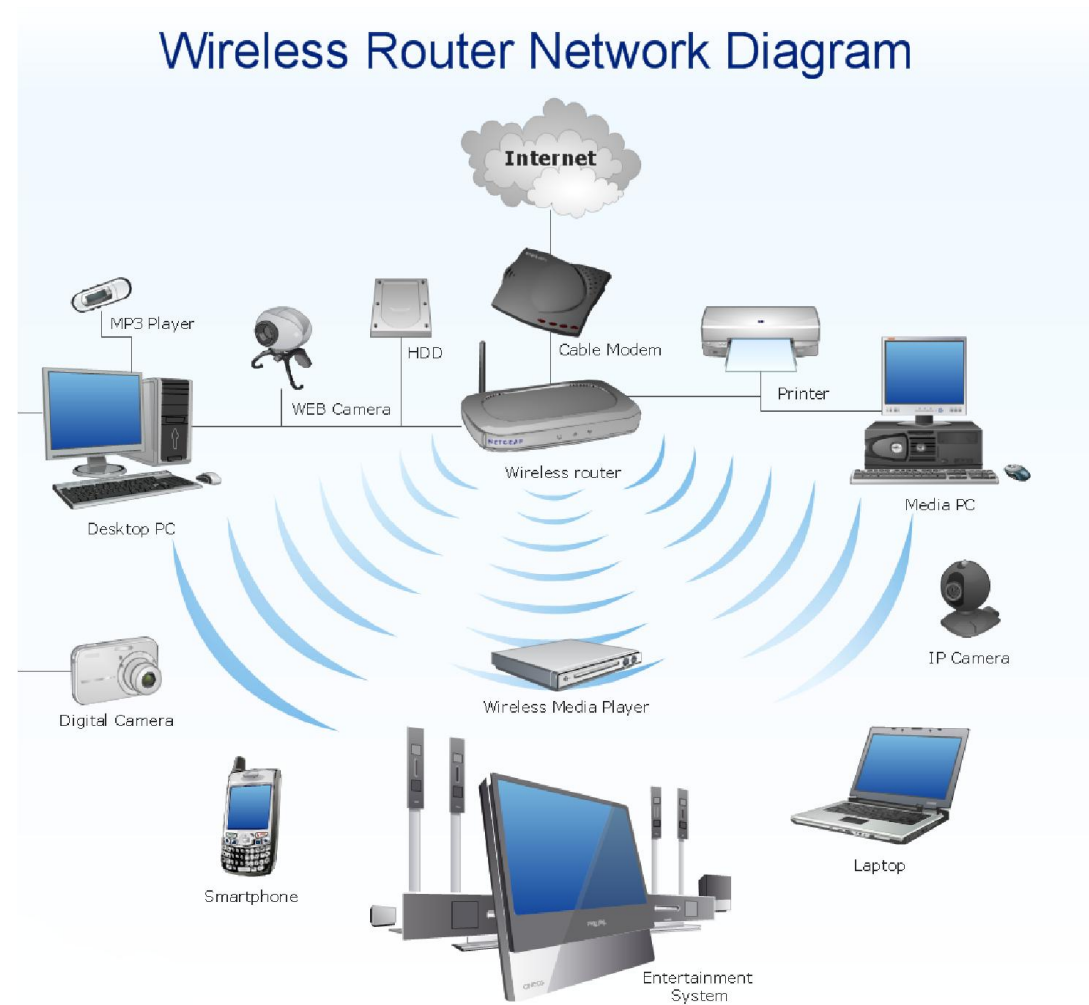
- Cellular Communication

- 1947: cellular concept first proposed at Bell labs.
- 1978: first cellular trial by AT&T.
- 1991: first GSM cellular service launched in Finland.
- 1996: 1st commercial CDMA cellular service launched in Finland.



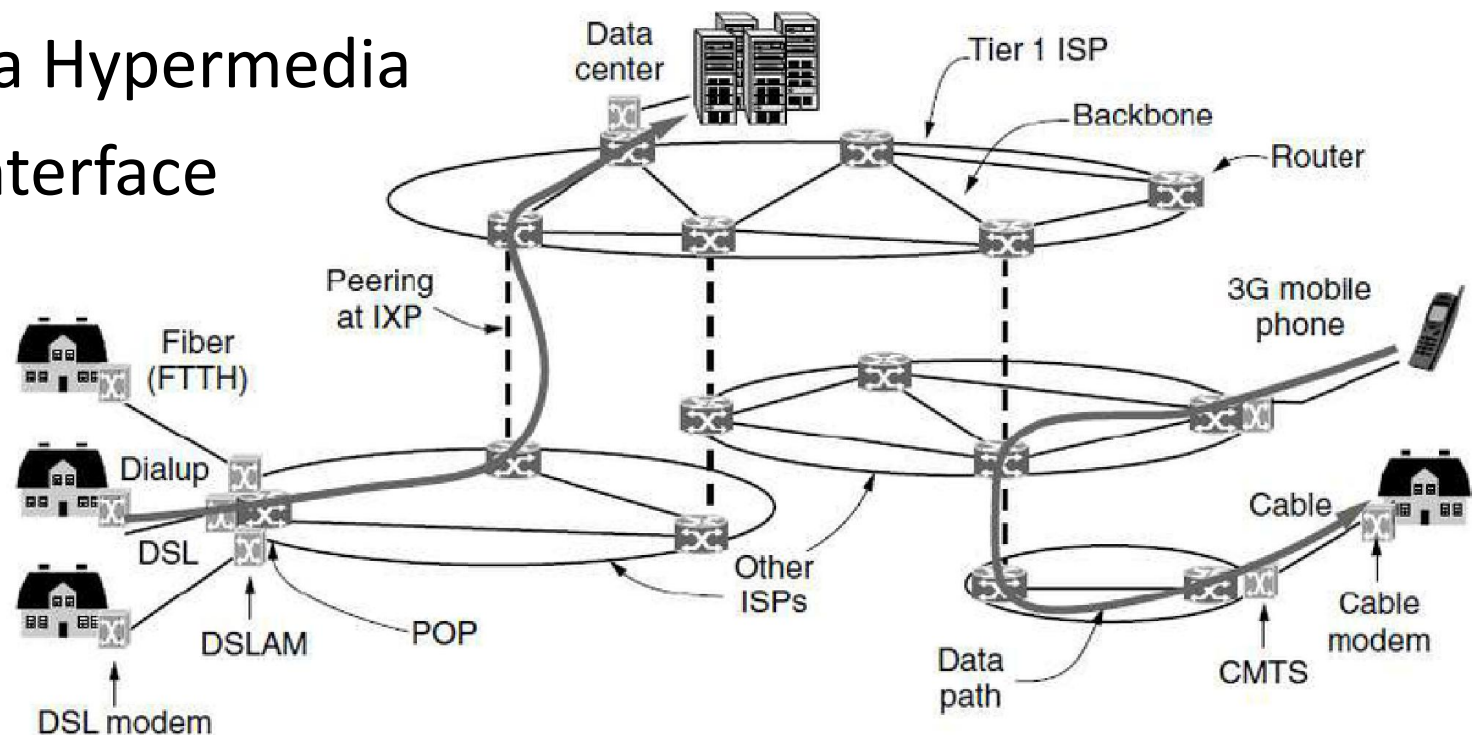
Communication system breakthroughs

- **Wireless LAN**
 - 1971: 1st wireless computer network
 - 1997- to date: *defined* a no. of different IEEE Wireless LAN standards.

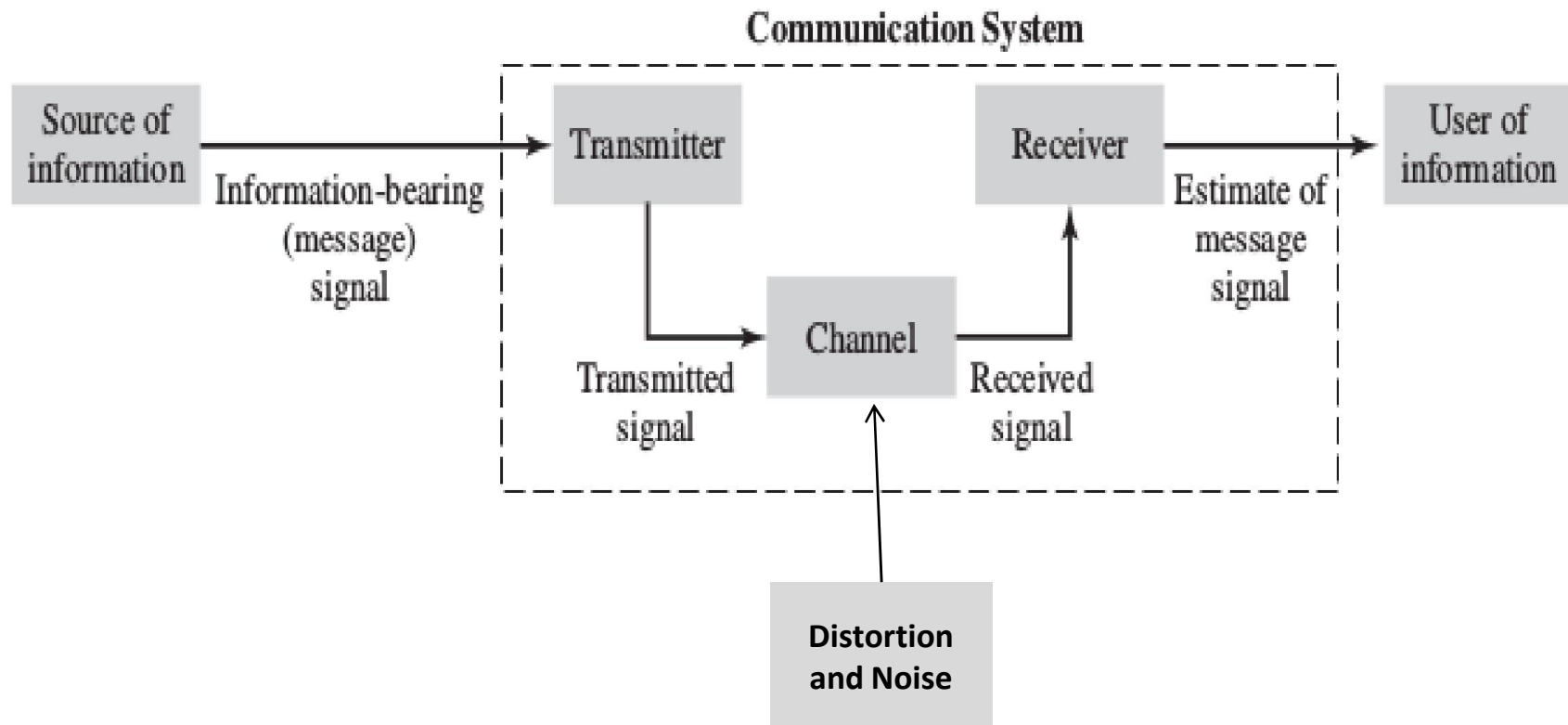


Communication system breakthroughs

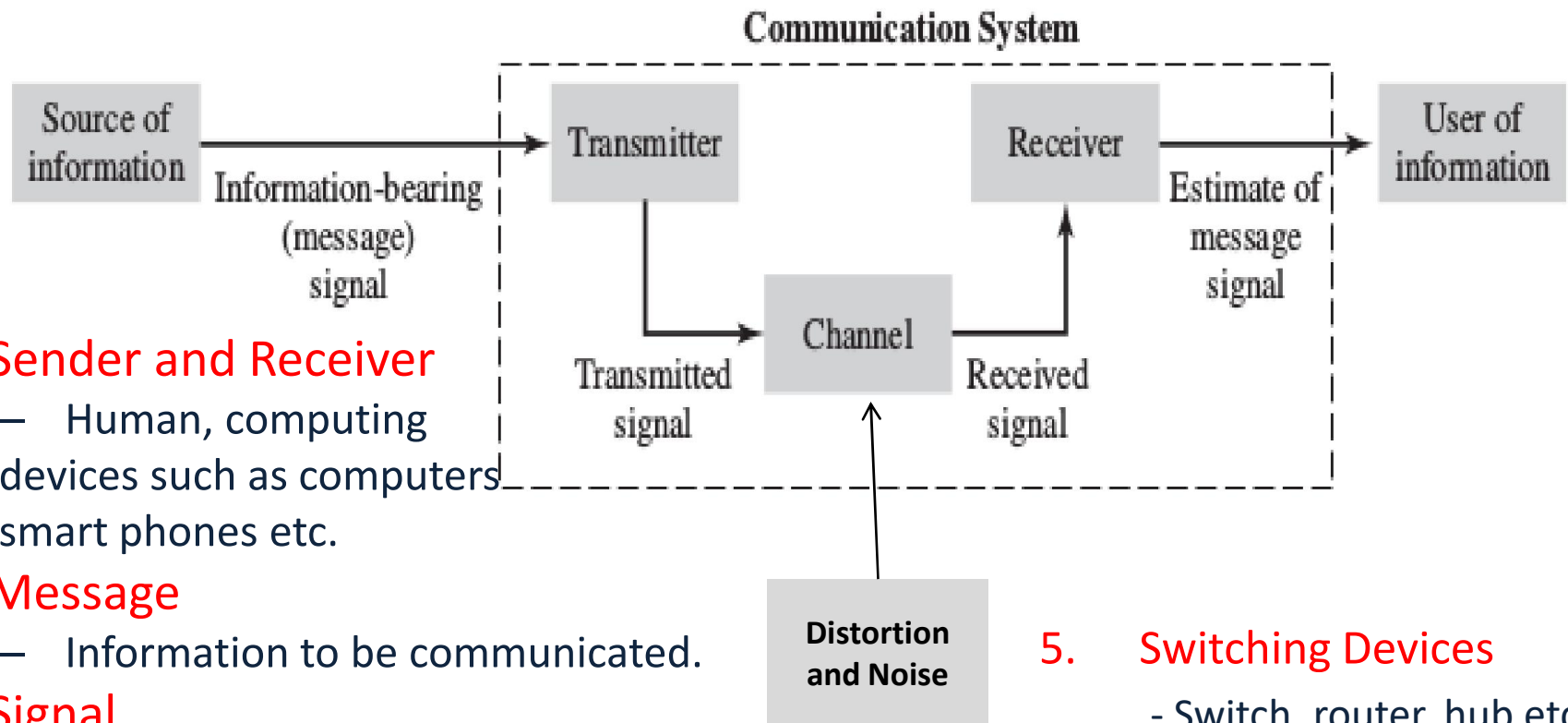
- Internet: Network of Networks
 - 1971: ARPANET put into service.
 - Later renamed the Internet, in 1985.
 - 1990: Tim Berners-Lee proposed a Hypermedia software interface to Internet WWW



Components of Communication systems



Components of Communication systems



1. Sender and Receiver

- Human, computing devices such as computers smart phones etc.

2. Message

- Information to be communicated.

3. Signal

- Electrical, Optical, Sound

4. Medium/Channel

- Air, Wire, Fiber, Vacuum

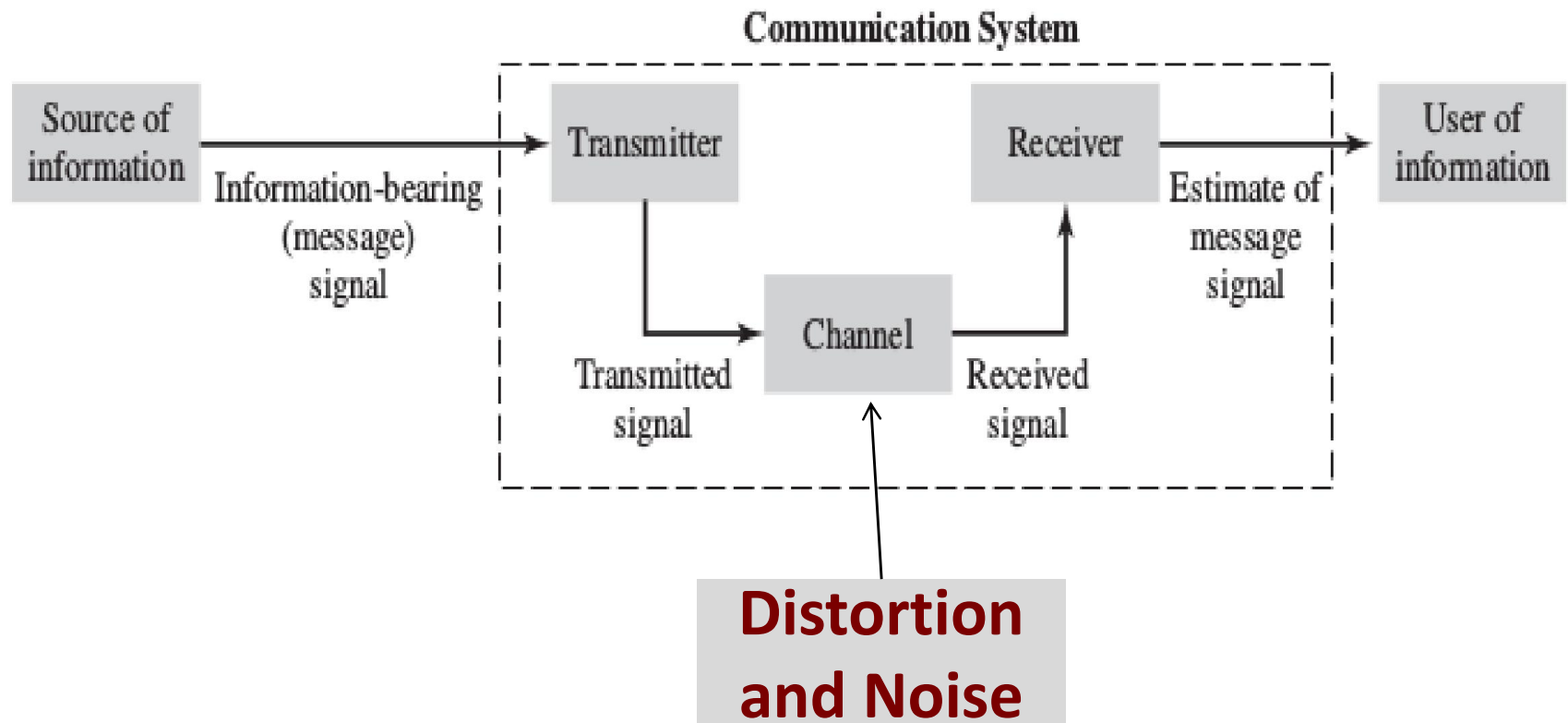
5. Switching Devices

- Switch, router, hub etc.

6. Protocols: Rules of Communication

- HTTP, SMTP, FTP etc.

Components of Communication systems



Main challenges

Challenges of Communication systems

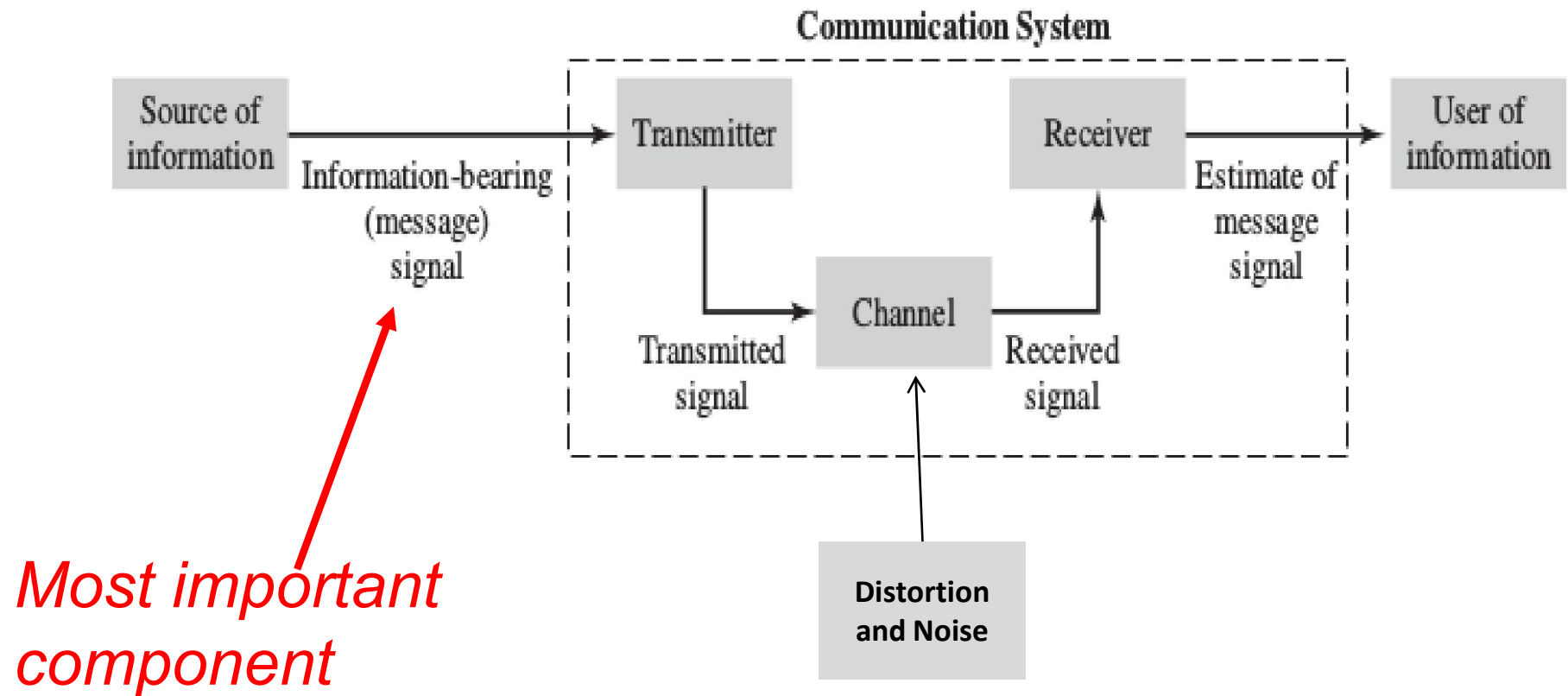
1. Distortion

- systematic undesirable changes in signals
- Linear or non-linear

2. Noise

- Unwanted signal that interfere with the transmitted signal
- Random signals from internal or external sources

Components of Communication systems



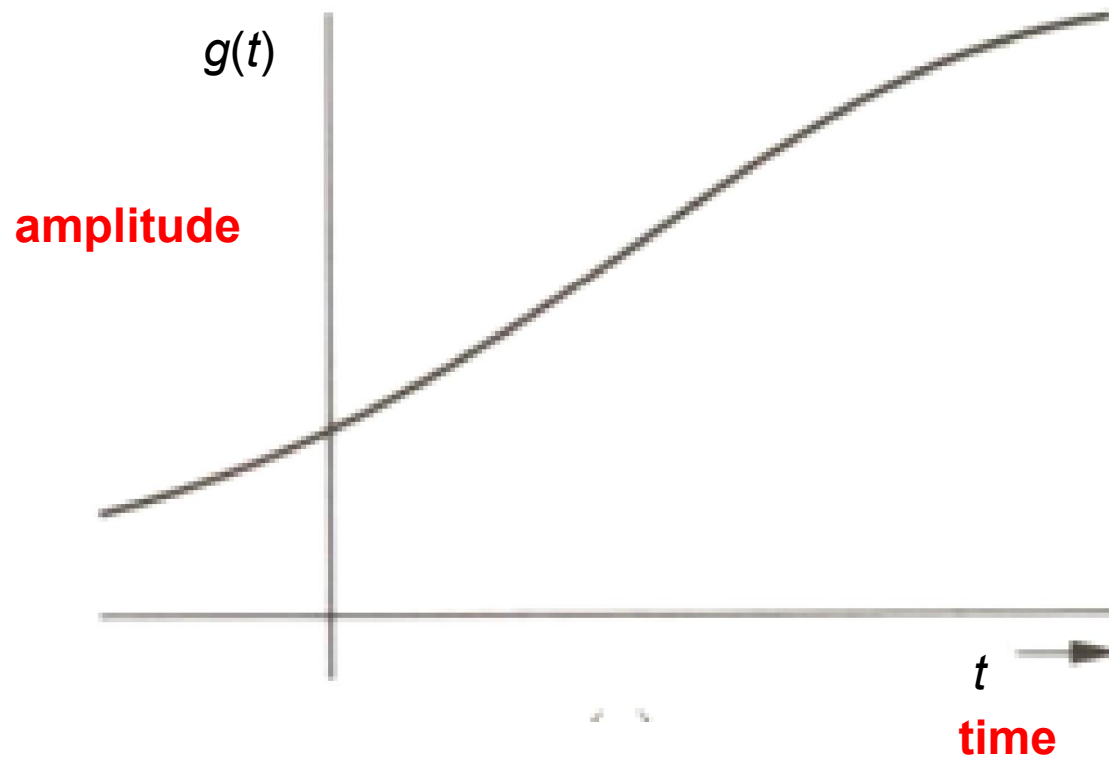
Messages/Signals: Definition

- A signal is a set of information or data.
- A signal is a function of independent variables that carry some information.
- A signal is a physical quantity that varies with time, space or any other independent variable by which information can be conveyed.

Example of Signals

- Voice signal
- Telephone or television signal
- Monthly sales figure
- Opening or closing stock prices
- Charge density over a surface
- In this course we deal with signals that are functions of time.

Signal representation: Time Domain



Classification of Signals

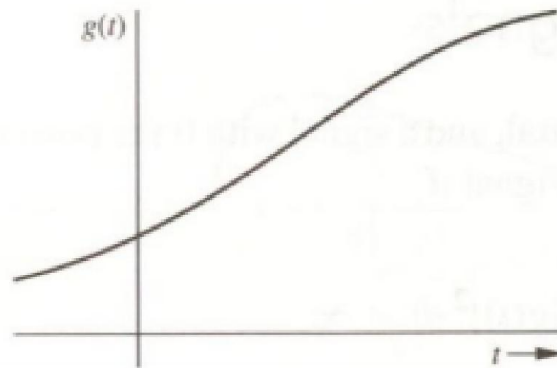
- Based on **continuity in time axis**
 - Continuous time
 - Discrete time
- Based on **continuity in amplitude axis**
 - Continuous amplitude
 - Discrete amplitude

Classification of Signals

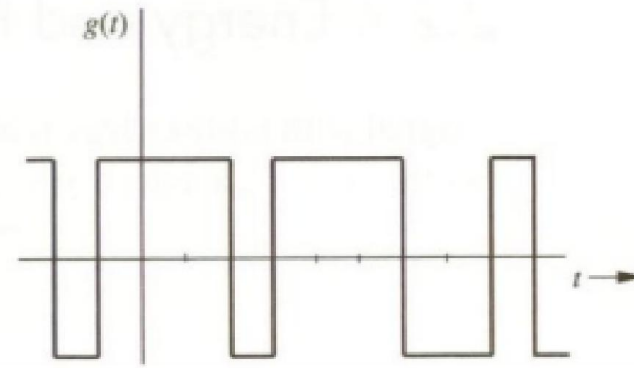
Continuous amplitude

Discrete amplitude

Continuous
time

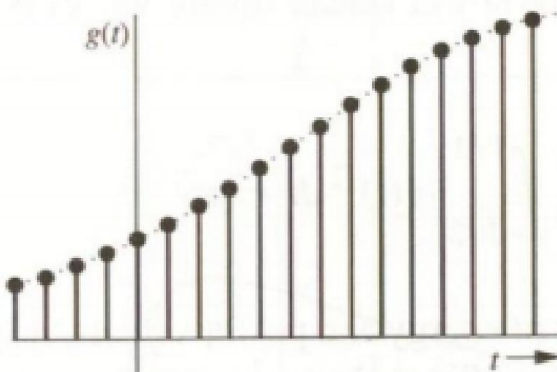


(a)

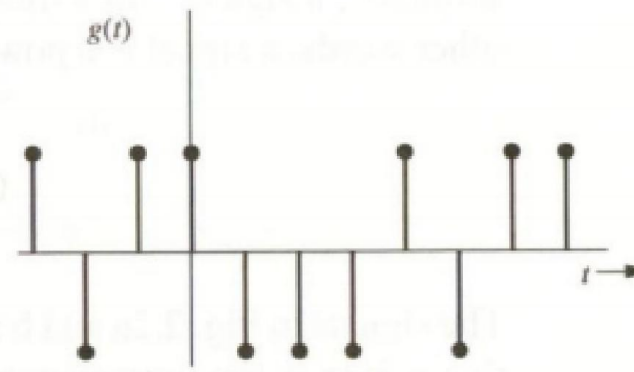


(b)

Discrete time



(c)



(d)

Analog and Digital Signal

Analog Signal

- Continuous amplitude, i.e., takes any value in a continuous range.
- May be both continuous and discrete time.

Digital Signal

- Discrete amplitude, i.e., amplitude can take only a finite number of values.
- Values need not be always integer.
- Not necessarily always binary, rather M-ary.
- May be both continuous and discrete time.